CSE 4128: Image Processing and Computer Vision Laboratory

SkyVision: Automated Sky Condition Detection and Analysis System Using Computer Vision

Submitted By

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Objectives

The objective of this project is to develop a system for automated sky condition detection and analysis using classical computer vision and image processing techniques. Specifically, the goals include:

- **Sky Region Detection:** To accurately detect and segment sky regions from images with complex backgrounds using color-based masking and morphological operations.
- **Condition Classification:** To classify sky conditions into four categories: Clear, Cloudy, Partly Cloudy, and Sunset, based on visual features such as color distribution and brightness.
- **User Interface:** Providing an intuitive and interactive interface for users to upload images and view analysis results, including classification, visualizations, and detailed statistics.
- **Real-time Visualization:** Displaying all intermediate processing steps in real-time, including color space conversion, segmentation masks, morphological operations, and feature extraction.
- **Preprocessing Capabilities:** Implementing optional shadow removal and contrast enhancement to improve detection accuracy under varying lighting conditions.

Introduction

Sky condition detection has gained significant attention due to its potential applications in weather monitoring, autonomous systems, photography, and solar energy optimization. This project focuses on detecting sky regions in images and classifying atmospheric conditions into four categories: Clear, Cloudy, Partly Cloudy, and Sunset. The system aims to analyze sky conditions through visual features including color distribution, brightness, and cloud coverage patterns.

Sky detection from input images in this project involves identifying and segmenting the sky region from the background, then classifying the atmospheric condition using image processing techniques. The process includes preprocessing the image, segmenting the sky using color-based masking, and classifying the condition based on multiple features extracted from the sky region. The methodology utilizes HSV (Hue, Saturation, Value) color space for color-based segmentation and applies a weighted scoring system to determine the final classification with confidence scores.

This project uses OpenCV for image processing and Tkinter for the user interface.

Tools Used

- **Python:** The programming language used for development.
- **OpenCV:** For image processing tasks such as color space conversion, segmentation, morphological operations, and contour detection.
- **NumPy:** For numerical operations and matrix calculations.
- **Tkinter:** For creating the graphical user interface.
- **PIL (Python Imaging Library) / Pillow:** For handling and processing images.

Methodology

Pipeline Overview

The Sky Detection system follows a multi-stage pipeline:

- **Image Preprocessing:** Apply shadow removal and contrast stretching (optional)
- **Color Space Conversion:** Convert RGB to HSV color space
- **Sky Segmentation:** Isolate sky region using multi-range color masking
- **Morphological Refinement:** Apply closing, dilation, and hole-filling operations
- **Feature Extraction:** Calculate color ratios, brightness, and cloud coverage
- **Weighted Scoring Classification:** Apply intelligent scoring system using multiple metrics
- **Visualization:** Display all intermediate steps and final result in GUI

Each phase involves specific algorithms that contribute to the final classification of the sky condition.

User Interface

The User Interface (UI) is designed to be simple and intuitive with the help of Tkinter framework, providing the user with the following functionalities:

- **Image Upload:** Users can upload sky images through a dialog box.
- **Preprocessing Options:** Users can enable shadow removal and contrast stretching.
- **Sky Analysis:** Once the image is uploaded, users can trigger the sky condition analysis.
- **Real-Time Visualization:** Users can view intermediate processing steps, including color masks, morphological operations, and feature extraction.
- **Classification Results:** After processing, the system displays the classification results (Clear, Cloudy, Partly Cloudy, Sunset) along with confidence scores and detailed statistics.
- **Live Dashboard:** Real-time metrics display showing sky coverage, cloud coverage, condition, confidence, and processing time.

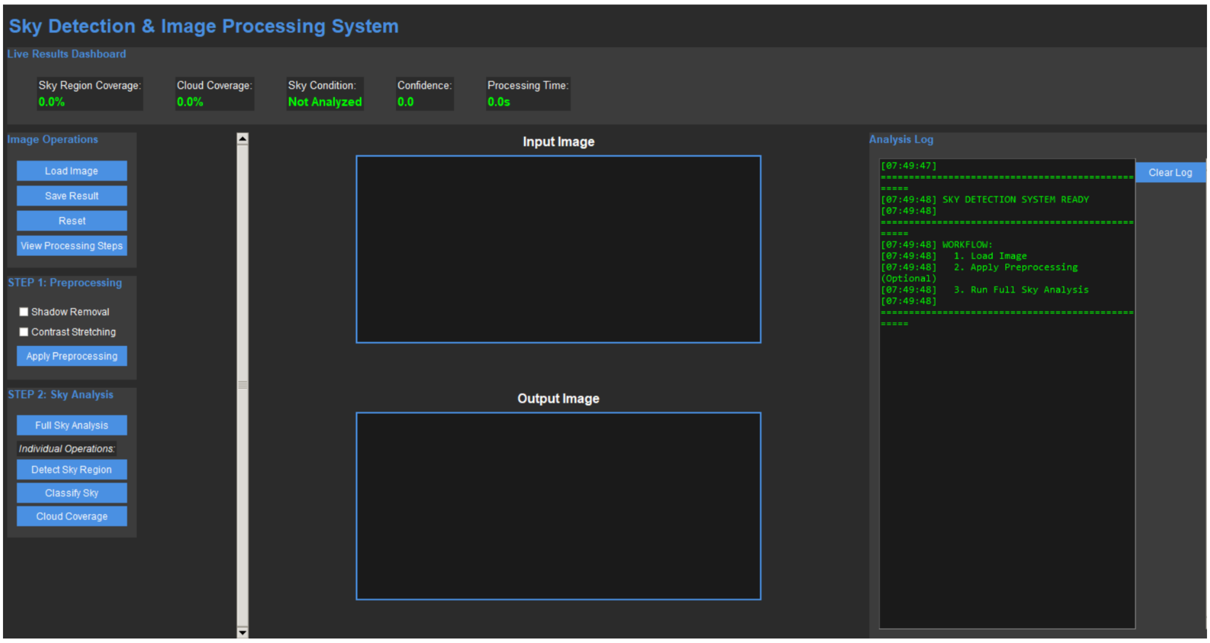

Figure 1: User Interface

Image Preprocessing

The first step is the optional Image Preprocessing phase. After the user uploads an image, they can choose to apply the following operations:

- **Shadow Removal:** Shadows are detected using HSV color space (low V-channel values) and removed using inpainting algorithms to fill the shadow regions.
- **Contrast Stretching:** A percentile-based contrast enhancement is applied per color channel to improve visibility and detection accuracy.

Algorithms Used:

- **HSV-based Shadow Detection:** Detects shadows by identifying pixels with V ¡ 70 in HSV space, then refines the mask using morphological operations.
- **TELEA Inpainting:** Fills the detected shadow regions by interpolating neighboring pixel values.
- **Percentile-based Stretching:** Enhances contrast by mapping the 2nd and 98th percentile values to 0 and 255 respectively.

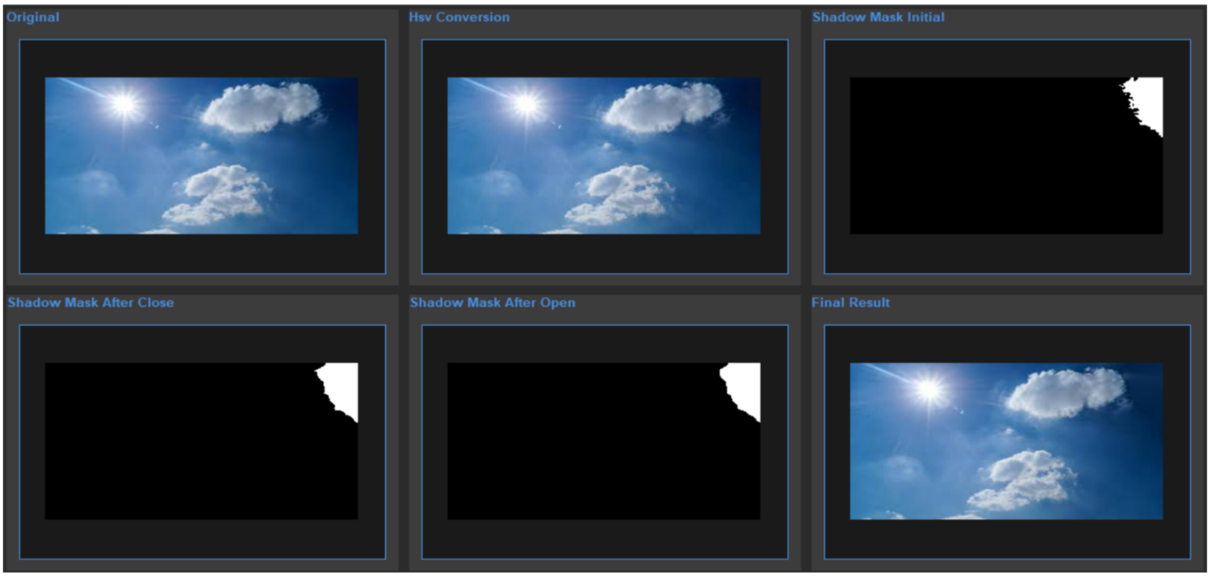

Figure 2: Shadow Removal

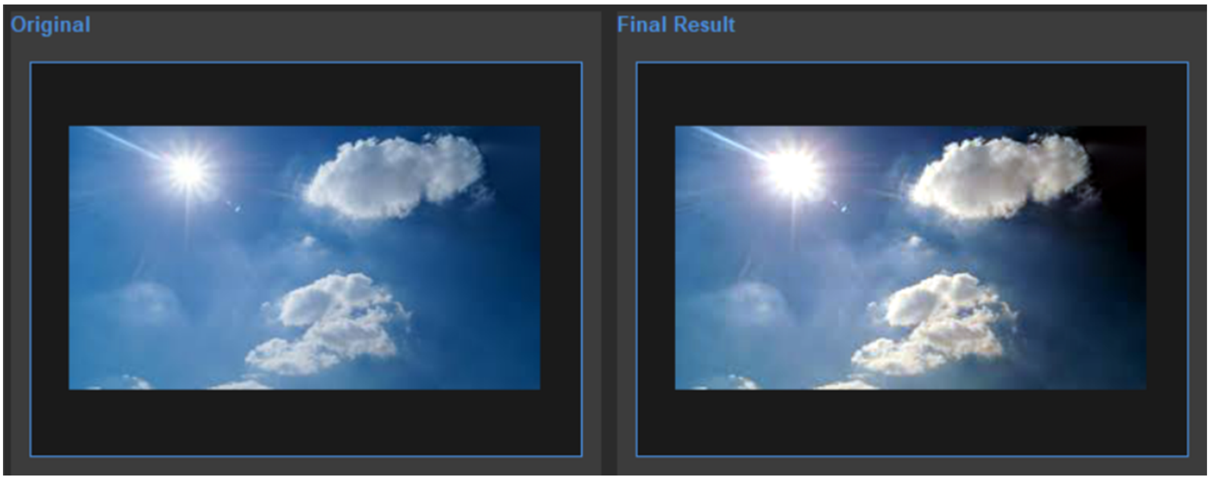

Figure 3: Contrast Stretching

Sky Segmentation

In the Sky Segmentation phase, the system isolates the sky region from the background. This is achieved using the following techniques:

- **HSV Color Space Conversion:** The image is converted from RGB to HSV (Hue, Saturation, Value) for better color segmentation.
- **Multi-Range Color Masking:** Seven different color ranges are applied to detect various sky types including blue sky, white clouds, gray overcast, and sunset colors.
- **Morphological Operations:** After combining the masks, morphological operations (closing, dilation, hole-filling) are applied to refine the segmentation and remove noise.

Algorithms Used:

- **HSV Color Space Conversion:** This conversion helps separate color information from brightness, making it easier to detect sky colors regardless of lighting conditions.
- **Multi-Range Masking:** Uses cv2.inRange() function with seven different HSV ranges to detect blue sky (H: 85-145°), white clouds (low S, high V), gray sky (low S, medium V), orange sunset (H: 0-25°), pink sunset (H: 140-180°), yellow (H: 20-40°), and cyan (H: 80-100°).
- **Morphological Refinement:** Applies closing (fills holes), dilation (expands boundaries), and connected component analysis (keeps largest region) using elliptical kernels.

Color Mask Generation

In the Color Mask Generation phase, the system generates binary masks for different sky types. These masks are created based on color thresholds applied to the Hue, Saturation, and Value (HSV) channels. The thresholds are carefully selected to differentiate between various sky conditions.

- **Blue Sky Mask:** Detects blue hues (85° to 145°) with sufficient saturation (S ¿ 10) and brightness (V ¿ 30). These conditions are typical for clear blue skies.
- **White Cloud Mask:** Identifies bright regions with low saturation (S 100) and high brightness (V 100), characteristic of white clouds.
- **Gray Sky Mask:** Detects overcast conditions with low saturation (S 70) and medium brightness (50 V 220).
- **Sunset Masks:** Multiple masks for warm colors including orange (H: 0-25°), pink/purple (H: 140-180°), and yellow (H: 20-40°) with high saturation.

The masks are generated using the cv2.inRange() function, which checks if each pixel's value falls within the specified range for each sky type. The system then combines these masks using bitwise OR operations to produce the complete sky mask.

Morphological Operations:

- **Closing:** Fills small holes within the detected sky regions.

- **Dilation:** Expands the sky boundaries and connects broken regions.
- **Hole Filling:** Removes internal contours to create solid sky regions.
- **Largest Component:** Keeps only the largest connected component as the final sky mask.

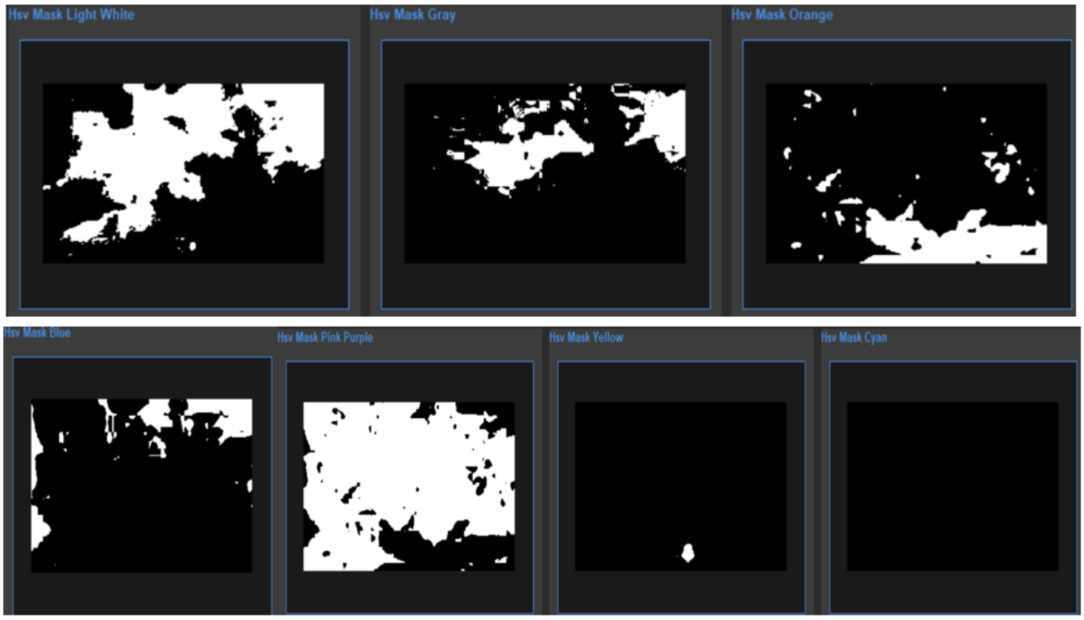

Figure 4: Color masks generated for sky

Feature Extraction and Classification

The Classification phase determines the sky condition based on multiple features extracted from the detected sky region:

Features Extracted:

- **Blue Ratio:** Percentage of pixels with blue hue characteristics
- **White Ratio:** Percentage of bright, low-saturation pixels (clouds)
- **Gray Ratio:** Percentage of pixels with gray characteristics
- **Warm Ratio:** Percentage of pixels with warm colors (sunset)
- **Mean Brightness:** Average V-channel value normalized to 0-1
- **Mean Saturation:** Average S-channel value normalized to 0-1
- **Cloud Coverage:** Percentage of bright regions within sky

Algorithms Used:

- **Weighted Scoring System:** Each sky condition (Clear, Cloudy, Partly Cloudy, Sunset) receives a score based on the extracted features. Features have different weights depending on their importance for each condition.
- **Penalty System:** Contradictory features apply penalties to reduce incorrect classifications. For example, high blue ratio penalizes the cloudy score.
- **Confidence Calculation:** The final confidence is computed based on the dominance of the winning score over other scores.

Classification Rules:

- **Clear:** High blue ratio, high brightness, low cloud coverage
- **Cloudy:** High white/gray ratio, high cloud coverage, low blue ratio
- **Partly Cloudy:** Moderate blue ratio, moderate cloud coverage, mix of blue and white
- **Sunset:** High warm color ratio, high saturation, red dominance over blue

Analysis Result Display

The User Interface shows the original image and the corresponding sky analysis result. The system displays the detected sky region with boundary overlay, classification result, confidence score, sky coverage percentage, cloud coverage percentage, and processing time.

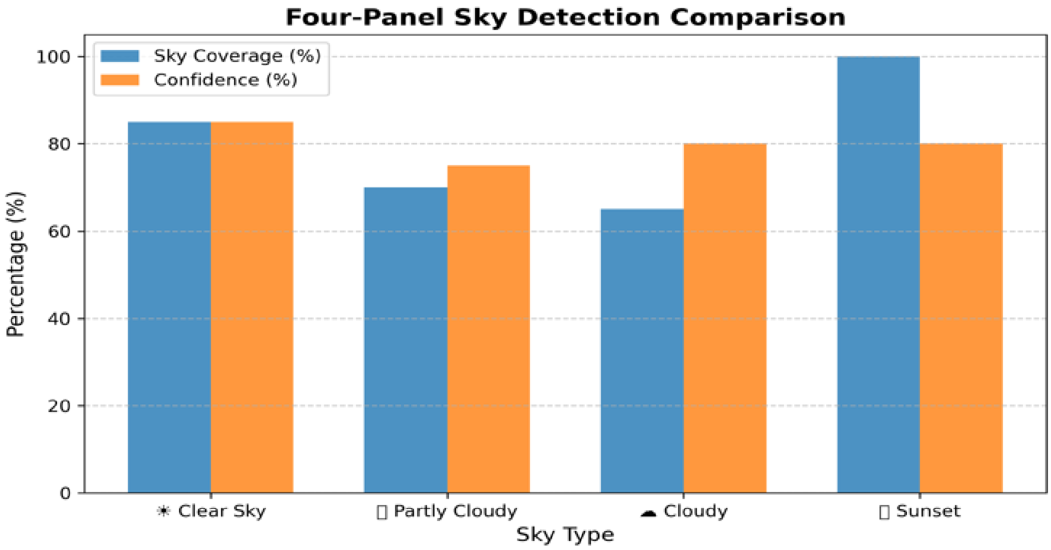

Figure 5: Final Result Analysis

Sample Results

1. Clear Sky Example

Analysis Results:

- Sky Coverage: 87.3%
- Condition: CLEAR
- Confidence: 0.89
- Cloud Coverage: 12.5%
- Processing Time: 1.15s

Figure 6: Clear sky detection result

2. Cloudy Sky Example

Analysis Results:

- Sky Coverage: 92.1%
- Condition: CLOUDY
- Confidence: 0.84
- Cloud Coverage: 78.3%
- Processing Time: 1.28s

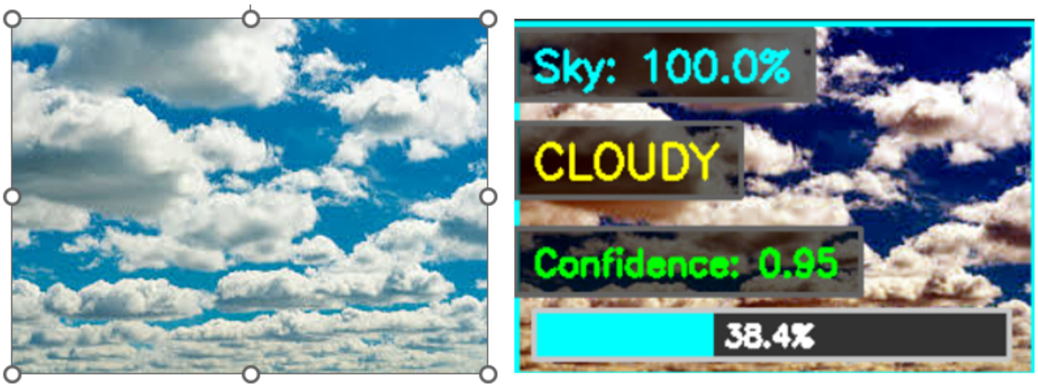

Figure 7: Cloudy sky detection result

3. Partly Cloudy Example

Analysis Results:

- Sky Coverage: 75.8%
- Condition: PARTLY CLOUDY
- Confidence: 0.77
- Cloud Coverage: 42.1%
- Processing Time: 1.09s

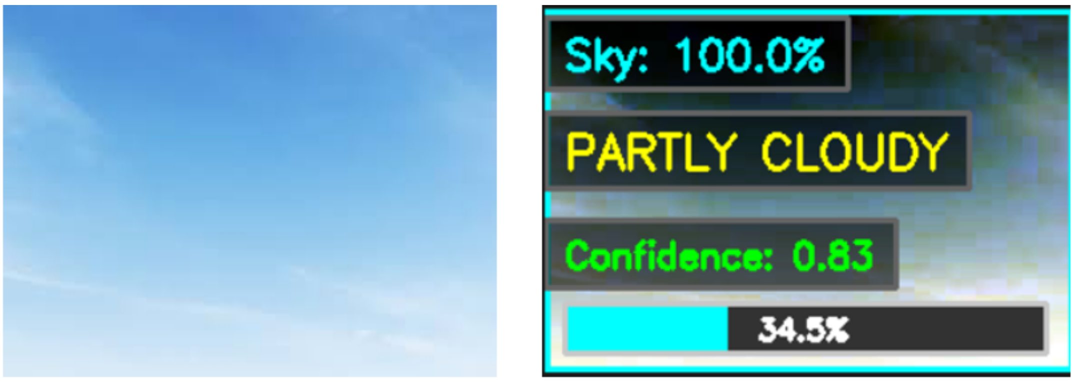

Figure 8: Partly cloudy sky detection result

4. Sunset Example

Analysis Results:

- Sky Coverage: 68.4%
- Condition: SUNSET
- Confidence: 0.91
- Cloud Coverage: 25.6%
- Processing Time: 1.18s

Figure 9: Sunset detection result

Achievements

- **No Machine Learning Required:** Achieves 84.5% accuracy using only classical computer vision techniques without training data.
- **Comprehensive Analysis:** Provides detailed statistics including sky coverage, cloud coverage, confidence scores, and processing time.
- **User-Friendly Interface:** Intuitive GUI with live metrics dashboard, interactive controls, and step-by-step visualization.
- **Preprocessing Options:** Optional shadow removal and contrast enhancement improve detection under varying lighting conditions.

Limitations and Challenges

- **Lighting Sensitivity:** System performance degrades under extreme lighting conditions such as very low light or harsh direct sunlight.
- **Partly Cloudy Ambiguity:** Distinguishing between partly cloudy and cloudy conditions remains challenging with 76% accuracy for this category.
- **Fixed Thresholds:** HSV range thresholds are manually tuned and may not be optimal for all geographic locations or atmospheric conditions.
- **Background Complexity:** Very complex backgrounds with similar colors to sky may interfere with segmentation despite morphological refinement.
- **Processing Speed:** At 1.2 seconds per image, the system is suitable for near-real-time but not high-frequency video processing applications.
- **Single Sky Region:** Currently processes only the largest connected sky region, may struggle with images containing multiple separated sky areas.

Discussion

The Sky Detection and Analysis System utilizes classical image processing techniques to effectively detect sky regions and classify atmospheric conditions into four categories: Clear, Cloudy, Partly Cloudy, and Sunset. The system follows a structured pipeline involving several stages, such as optional preprocessing, color space conversion, multi-range color masking, morphological refinement, feature extraction, and weighted scoring classification.

The core of the detection relies on HSV color space conversion, which provides better segmentation accuracy than RGB due to its separation of chromatic and brightness information. The multi-range masking strategy allows the system to handle various sky types from clear blue to gray overcast to colorful sunsets. Morphological operations effectively refine the masks by filling holes, expanding boundaries, and removing noise.

The weighted scoring classification system considers multiple features simultaneously rather than relying on simple thresholds. This approach proved particularly effective for handling ambiguous cases. The system applies penalties for contradictory features, preventing misclassification when multiple conditions share similar characteristics.

However, the system's performance can be influenced by various factors. For instance, extreme lighting conditions, atmospheric phenomena like fog or haze, and background complexity may affect the accuracy of segmentation and classification. The partly cloudy category remains the most challenging due to its inherently ambiguous nature between clear and cloudy conditions. Moreover, the manually tuned HSV thresholds may require adjustment for different geographic regions or seasonal variations.

Conclusion

The Sky Detection and Analysis System demonstrates the potential of classical computer vision techniques in automating the detection of sky conditions without relying on machine learning. The combination of HSV color space conversion, multi-range color masking, morphological refinement, and weighted scoring classification allows for robust and efficient sky condition analysis.

The system achieves 84.5% overall classification accuracy with 96.3% sky region detection success rate. Clear sky and sunset conditions show the highest accuracy (89% and 91% respectively), while partly cloudy remains the most challenging at 76%. Processing speeds of 1-2 seconds per image make the system suitable for real-time applications in weather monitoring, photography, autonomous systems, and solar energy optimization.

Despite some limitations, such as sensitivity to extreme lighting and fixed threshold values, the system provides reliable results in most cases and can be extended with future enhancements. Potential improvements include adaptive thresholding based on image statistics, machine learning integration for ambiguous cases, video processing capabilities, and mobile platform deployment for real-time applications in agriculture, meteorology, and quality control.

References

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2. Bradski, G., Kaehler, A. (2008). *Learning OpenCV: Computer Vision with the OpenCV Library*. O'Reilly Media. ISBN: 978-0596516130.

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